

$$\begin{aligned}
 \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} &= \begin{bmatrix} -5x_2 - 8x_4 - x_5 \\ x_2 \\ 7x_4 - 4x_5 \\ x_4 \\ x_5 \\ 0 \end{bmatrix} \\
 \text{param: } x_2 &= p, \quad x_4 = t, \quad x_5 = q
 \end{aligned}$$

$$\begin{aligned}
 \sim p \begin{bmatrix} -5 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} &+ t \begin{bmatrix} -8 \\ 0 \\ 7 \\ 0 \\ 0 \end{bmatrix} + q \begin{bmatrix} -1 \\ 0 \\ -4 \\ 0 \\ 0 \end{bmatrix}
 \end{aligned}$$

$$\begin{aligned}
 \sin 2x &= \frac{3}{5} \\
 \sin(x+x)
 \end{aligned}$$

$$\sin x = ?$$

$$\cos 2x = 2 \cos^2 x - 1$$

$$\begin{aligned}
 \frac{4}{5} &= 2 \cos^2 x - 1 \\
 \frac{9}{10}
 \end{aligned}$$

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} = \begin{bmatrix} -5x_2 - 8x_4 - x_5 \\ x_2 \\ 7x_4 - 4x_5 \\ x_4 \\ x_5 \\ 0 \end{bmatrix}$$

param: $x_2 = p$ $x_4 = t$ $x_5 = q$

$$\sim \begin{bmatrix} -5 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} -8 \\ 0 \\ 7 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} -1 \\ 0 \\ -4 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\sin 2x = \frac{3}{5}$$

$$\sin(x+x)$$

$$\sin x = ?$$

$$\cos 2x = 2\cos^2 x - 1$$

$$\frac{4}{5} = 2\cos^2 x - 1$$

$$\frac{9}{10}$$

vector

matrix

$$x_1 = 5x_3$$

$$x_2 = -2x_3$$

$$x_3 = x_3$$

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 5x_3 \\ -2x_3 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} 5 \\ -2 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 1x_1 & 5x_2 & 2x_3 & -6x_4 & 9x_5 & 0x_6 \\ & & 1 & -7 & 4 & 8 \\ & & & & & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$1.5x_2$$

$$\begin{bmatrix} 1 & 5 & 2 & -6 & 9 & 0 \\ & & 1 & -7 & 4 & 0 \\ & & & & & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 5 & 0 & 8 & 1 & 0 \\ & & 1 & -7 & 4 & 0 \\ & & & & & 1 \end{bmatrix}$$

$$x_3 = 7x_4 - 4x_5$$

$$x_1 = -5x_2 - 8x_4 - x_5$$

$$x_6 = 0$$

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

vector

matrix

$$x_1 = 5x_3$$

$$x_2 = -2x_3$$

$$x_3 = x_3$$

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 5x_3 \\ -2x_3 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} 5 \\ -2 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 1x_1 & 5x_2 & 2x_3 & -6x_4 & 9x_5 & 0x_6 \\ & & 1 & -7 & 4 & 8 \\ & & & & & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$1.5x_2$$

$$\begin{bmatrix} 1 & 5 & 2 & -6 & 9 & 0 \\ & & 1 & -7 & 4 & 0 \\ & & & & & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 5 & 0 & 8 & 1 & 0 \\ & & 1 & -7 & 4 & 0 \\ & & & & & 1 \end{bmatrix}$$

$$x_3 = 7x_4 - 4x_5$$

$$x_1 = -5x_2 - 8x_4 - x_5$$

$$x_6 = 0$$

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

1.4 to

$$x_1 \begin{bmatrix} 3 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + x_3 \begin{bmatrix} -5 \\ 4 \end{bmatrix} = \begin{bmatrix} 9 \\ 0 \end{bmatrix} \quad \text{vector}$$

$$\begin{bmatrix} 3 & 1 & -5 \\ 0 & 1 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 9 \\ 0 \end{bmatrix} \quad \text{matrix}$$

1.4 to

$$\begin{bmatrix} 3 & -6 & | & 0 \\ 2 & 6 & | & 4 \\ 1 & 1 & | & 4 \end{bmatrix} \quad \begin{bmatrix} 0 & -9 & | & -12 \\ 0 & 4 & | & -4 \\ 1 & 1 & | & 4 \end{bmatrix} \quad \text{5/5}$$

$$\begin{bmatrix} 1 & 1 & 4 \\ 0 & 1 & -1 \\ 0 & 3 & 4 \end{bmatrix} \quad \begin{bmatrix} 1 & 1 & 4 \\ 0 & 1 & -1 \\ 0 & 0 & 7 \end{bmatrix} \quad \phi$$

1.5 to

$$\begin{bmatrix} 1 & 3 & 1 \\ -4 & -9 & 2 \\ 0 & -3 & -6 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & 1 \\ 0 & 3 & 6 \\ 0 & -3 & -6 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 3 & 1 \\ 0 & 3 & 6 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -5 \\ 0 & 3 & 6 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

1.4 to

$$x_1 \begin{bmatrix} 3 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + x_3 \begin{bmatrix} -5 \\ 4 \end{bmatrix} = \begin{bmatrix} 9 \\ 0 \end{bmatrix} \quad \text{vector}$$

$$\begin{bmatrix} 3 & 1 & -5 \\ 0 & 1 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 9 \\ 0 \end{bmatrix} \quad \text{matrix}$$

1.4 to

$$\begin{bmatrix} 3 & 1 & -5 & | & 9 \\ 0 & 1 & 4 & | & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & -1 & -12 & | & -9 \\ 0 & 1 & 4 & | & 0 \end{bmatrix} \quad \text{5/5}$$

$$\begin{bmatrix} 1 & 1 & 4 \\ 0 & 1 & -1 \\ 0 & 3 & 4 \end{bmatrix} \quad \begin{bmatrix} 1 & 1 & 4 \\ 0 & 1 & -1 \\ 0 & 0 & 8 \end{bmatrix} \quad \phi$$

1.5 to

$$\begin{bmatrix} 1 & 3 & 1 \\ -4 & -9 & 2 \\ 0 & -3 & -6 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & 1 \\ 0 & 3 & 6 \\ 0 & -3 & -6 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 3 & 1 \\ 0 & 3 & 6 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -5 \\ 0 & 3 & 6 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$6) v_1 = 1, v_2 = 2$$

$$1 \begin{bmatrix} 2 \\ 1 \\ -6 \end{bmatrix} + 1 \begin{bmatrix} -5 \\ 3 \\ 0 \end{bmatrix} = \begin{bmatrix} -3 \\ 4 \\ -6 \end{bmatrix}$$

$$\left[\begin{array}{cc|c} 1 & -2 & 4 \\ 4 & -3 & 1 \\ -2 & 7 & h \end{array} \right]$$

$$\begin{cases} x_1 - 2x_2 = 4 \\ 4x_1 - 3x_2 = 1 \\ -2x_1 + 7x_2 = ? \end{cases}$$

$$-1.3 + 1.2$$

$$x_2 = -3$$

$$x_1 = -2 \checkmark$$

$$1 \quad -2 \quad 4$$

$$0 \quad 5 \quad -15$$

$$h = 4 - 21 = -17$$

~~$$\begin{bmatrix} 2 \\ 1 \\ -6 \end{bmatrix} \begin{bmatrix} -5 \\ 3 \\ 0 \end{bmatrix}$$~~

$$\text{len}(V) = 2$$

$$\text{len}(W) = 3$$

$$1.9 + 1.2$$

05504

$$\begin{bmatrix} 6 \cdot 2 - 3 \cdot 5 \\ -4 \cdot 2 + 3 \cdot 3 \\ 7 \cdot 2 - 3 \cdot 6 \end{bmatrix} = \begin{bmatrix} -3 \\ +1 \\ -4 \end{bmatrix}$$

$$1.9 \neq 3$$

$$1.9 \neq 4$$

$$\begin{bmatrix} 8 + 3 - 4 \\ 5 + 1 + 2 \end{bmatrix} \approx \begin{bmatrix} 7 \\ 8 \end{bmatrix}$$

$$6) v_1 = 1, v_2 = 2$$

$$1 \begin{bmatrix} 2 \\ 1 \\ -6 \end{bmatrix} + 1 \begin{bmatrix} -5 \\ 3 \\ 0 \end{bmatrix} = \begin{bmatrix} -3 \\ 4 \\ -6 \end{bmatrix}$$

$$\left[\begin{array}{cc|c} 1 & -2 & 4 \\ 4 & -3 & 1 \\ -2 & 7 & h \end{array} \right]$$

$$\begin{cases} x_1 - 2x_2 = 4 \\ 4x_1 - 3x_2 = 1 \\ -2x_1 + 7x_2 = ? \end{cases}$$

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~~$$\begin{bmatrix} 2 \\ 1 \\ -6 \end{bmatrix} \begin{bmatrix} -5 \\ 3 \\ 0 \end{bmatrix}$$~~

$$\text{len}(V) = 2$$

~~$$\text{len}(C) = 1.9 + 1.2$$~~

$$\text{len}(COW) = 3$$

05504

$$\begin{bmatrix} 6 \cdot 2 - 3 \cdot 5 \\ -4 \cdot 2 + 3 \cdot 3 \\ 7 \cdot 2 - 3 \cdot 6 \end{bmatrix} = \begin{bmatrix} -3 \\ +1 \\ -4 \end{bmatrix}$$

$$1.9 \neq 3$$

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$$\begin{bmatrix} 8 + 3 - 4 \\ 5 + 1 + 2 \end{bmatrix} \approx \begin{bmatrix} 7 \\ 8 \end{bmatrix}$$

$$\begin{bmatrix} 6 & -3 & 1 \\ -1 & 4 & -2 \\ 5 & 0 & -5 \end{bmatrix}$$

$$\begin{aligned} x_1 &= 8 \\ x_2 &= 42 \end{aligned}$$

$$\begin{bmatrix} 1 & -4 & 7 \\ 6 & -3 & 1 \\ 5 & 0 & 5 \end{bmatrix} \quad \begin{bmatrix} 1 & -4 & 7 \\ 0 & 21 & -41 \\ 0 & 20 & -30 \end{bmatrix}$$

$$x_2 = -\frac{41}{21} \neq \frac{30}{20} \quad \emptyset$$

1.3 + 13

$$\begin{array}{ccc|c} 1 & -4 & 2 & 3 \\ 0 & 3 & 5 & -7 \\ -2 & 8 & -4 & -3 \end{array}$$

← if it was -6
it would be possible

1.3 + 15?

$$1) v_1 = \begin{bmatrix} 7 \\ 1 \\ -6 \end{bmatrix}$$

$$v_2 = \begin{bmatrix} -5 \\ 3 \\ 0 \end{bmatrix}$$

$$1) v_1 = 0; v_2 = 0$$

$$0 \begin{bmatrix} 7 \\ 1 \\ -6 \end{bmatrix} + 0 \begin{bmatrix} -5 \\ 3 \\ 0 \end{bmatrix} = 0$$

$$2) v_1 = 1; v_2 = 0$$

$$1 \begin{bmatrix} 7 \\ 1 \\ -6 \end{bmatrix} + 0 \begin{bmatrix} -5 \\ 3 \\ 0 \end{bmatrix} = \begin{bmatrix} 7 \\ 1 \\ -6 \end{bmatrix}$$

$$3) v_1 = 0; v_2 = 1$$

$$0 \begin{bmatrix} 7 \\ 1 \\ -6 \end{bmatrix} + 1 \begin{bmatrix} -5 \\ 3 \\ 0 \end{bmatrix} = \begin{bmatrix} -5 \\ 3 \\ 0 \end{bmatrix}$$

$$4) v_1 = 1; v_2 = 1$$

$$1 \begin{bmatrix} 7 \\ 1 \\ -6 \end{bmatrix} + 1 \begin{bmatrix} -5 \\ 3 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ -6 \end{bmatrix}$$

$$\begin{bmatrix} 6 & -3 & 1 \\ -1 & 4 & -2 \\ 5 & 0 & -5 \end{bmatrix}$$

$$\begin{aligned} x_1 &= 8 \\ x_2 &= 42 \end{aligned}$$

$$\begin{bmatrix} 1 & -4 & 7 \\ 6 & -3 & 1 \\ 5 & 0 & 5 \end{bmatrix} \quad \begin{bmatrix} 1 & -4 & 7 \\ 0 & 21 & -41 \\ 0 & 20 & -30 \end{bmatrix}$$

$$x_2 = -\frac{41}{21} \neq \frac{30}{20} \quad \emptyset$$

1.3 + 13

$$\begin{array}{ccc|c} 1 & -4 & 2 & 3 \\ 0 & 3 & 5 & -7 \\ -2 & 8 & -4 & -3 \end{array}$$

← if it was -6
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1.3 + 15?

$$1) v_1 = \begin{bmatrix} 7 \\ 1 \\ -6 \end{bmatrix}$$

$$v_2 = \begin{bmatrix} -5 \\ 3 \\ 0 \end{bmatrix}$$

$$1) v_1 = 0; v_2 = 0$$

$$0 \begin{bmatrix} 7 \\ 1 \\ -6 \end{bmatrix} + 0 \begin{bmatrix} -5 \\ 3 \\ 0 \end{bmatrix} = 0$$

$$2) v_1 = 1; v_2 = 0$$

$$1 \begin{bmatrix} 7 \\ 1 \\ -6 \end{bmatrix} + 0 \begin{bmatrix} -5 \\ 3 \\ 0 \end{bmatrix} = \begin{bmatrix} 7 \\ 1 \\ -6 \end{bmatrix}$$

$$3) v_1 = 0; v_2 = 1$$

$$0 \begin{bmatrix} 7 \\ 1 \\ -6 \end{bmatrix} + 1 \begin{bmatrix} -5 \\ 3 \\ 0 \end{bmatrix} = \begin{bmatrix} -5 \\ 3 \\ 0 \end{bmatrix}$$

$$4) v_1 = 1; v_2 = 1$$

$$1 \begin{bmatrix} 7 \\ 1 \\ -6 \end{bmatrix} + 1 \begin{bmatrix} -5 \\ 3 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ -6 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 & 4 & 7 \\ 3 & 9 & 7 & 6 \end{bmatrix} \quad -3R_1$$

1,2+7

$$\begin{bmatrix} 1 & 3 & 4 & 7 \\ 0 & 0 & -5\frac{1}{3} & -15 \end{bmatrix}$$

$$0x_1 + 0x_2 - 5\frac{1}{3}x_3 = -15$$

$$x_3 = +\frac{15}{5}$$

$$x_3 = 3$$

$$x_2 \rightarrow \text{free}$$

$$x_1 = 7 - 3x_2$$

$$\begin{bmatrix} 1 & -3 & -1 & 0 & -2 \\ 0 & 1 & 0 & -4 & 1 \\ 0 & 0 & 1 & 9 & 4 \end{bmatrix}$$

XGL

1,2+13

$$x_1 + 3x_2 - x_3 = -2$$

$$x_2 - 4x_4 = 1$$

$$x_3 + 9x_4 = 4$$

$$x_1 - 3(1 + 4x_4) - (4 - 9x_4) = -2$$

$$x_2 = 1 + 4x_4$$

$$x_3 = 4 - 9x_4$$

-?

$$\begin{bmatrix} -1 & -3 \\ 2 & -1 \end{bmatrix} \sim \begin{bmatrix} -1 & 6 \\ 2 & 2 \end{bmatrix}$$

1,3+1

1,3+5

$$\begin{bmatrix} 1 & 3 & 4 & 7 \\ 3 & 9 & 7 & 6 \end{bmatrix} \quad -3R_1$$

1,2+7

$$\begin{bmatrix} 1 & 3 & 4 & 7 \\ 0 & 0 & -5\frac{1}{3} & -15 \end{bmatrix}$$

$$0x_1 + 0x_2 - 5\frac{1}{3}x_3 = -15$$

$$x_3 = +\frac{15}{5\frac{1}{3}}$$

$$x_3 = 3$$

$$x_2 \rightarrow \text{free}$$

$$x_1 = 7 - 3x_2$$

$$\begin{bmatrix} 1 & -3 & -1 & 0 & -2 \\ 0 & 1 & \frac{2}{9} & -4 & 1 \\ 0 & 0 & 1 & 9\frac{1}{9} & 9 \end{bmatrix}$$

XGL

1,2+13

$$x_1 + 3x_2 - x_3 = -2$$

$$x_2 - 4x_4 = 1$$

$$x_3 + 9x_4 = 9$$

$$x_1 - 3(1 + 4x_4) - (9 - 9x_4) = -2$$

$$x_2 = 1 + 4x_4$$

$$x_3 = 9 - 9x_4$$

-?

$$\begin{bmatrix} -1 & -3 \\ 2 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 6 \\ 2 & 2 \end{bmatrix}$$

1,3+1

1,3+5

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \end{bmatrix} \quad R_1 - 2R_2$$

$$\begin{bmatrix} 1 & 0 & -1 & -2 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$1, 2, 4 \quad \begin{bmatrix} 1 & 3 & 5 & 7 \\ 3 & 5 & 7 & 9 \\ 5 & 7 & 9 & 1 \end{bmatrix} \quad \begin{array}{l} -3R_1 \\ -5R_1 \end{array}$$

$$\begin{bmatrix} 1 & 3 & 5 & 7 \\ 0 & -4 & -8 & -12 \\ 0 & -8 & -16 & -34 \end{bmatrix} \quad -2R_2$$

$$\begin{bmatrix} 1 & 3 & 5 & 7 \\ 0 & -4 & -8 & -12 \\ 0 & 0 & 0 & -8 \end{bmatrix}$$

~~0~~

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \end{bmatrix} \quad R_1 - 2R_2$$

$$\begin{bmatrix} 1 & 0 & -1 & -2 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$1, 2, 4 \quad \begin{bmatrix} 1 & 3 & 5 & 7 \\ 3 & 5 & 7 & 9 \\ 5 & 7 & 9 & 1 \end{bmatrix} \quad \begin{array}{l} -3R_1 \\ -5R_1 \end{array}$$

$$\begin{bmatrix} 1 & 3 & 5 & 7 \\ 0 & -4 & -8 & -12 \\ 0 & -8 & -16 & -34 \end{bmatrix} \quad -2R_2$$

$$\begin{bmatrix} 1 & 3 & 5 & 7 \\ 0 & -4 & -8 & -12 \\ 0 & 0 & 0 & -8 \end{bmatrix}$$

~~0~~

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 4 & 5 & 6 & 7 \\ 6 & 7 & 8 & 9 \end{bmatrix} \begin{matrix} -R_1 \\ -R_2 \end{matrix}$$

1.2+3

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 3 & 3 & 3 & 3 \\ 2 & 2 & 2 & 2 \end{bmatrix}$$

$$x_1 + x_2 + x_3 = 1$$

$$x_1 + 2x_2 + 3x_3 = 9$$

$$x_2 + 2x_3 = 3; \quad x_1 = 1$$

not solvable.

✓

$$R_2 - 4R_1 - 3 \cdot (x_2 + 2x_3) \Rightarrow 2x_2 = -2; \quad x_2 = -1$$

$$x_3 =$$

Re-calc.

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 9 & 5 & 6 & 7 \\ 6 & 2 & 8 & 9 \end{bmatrix} \begin{matrix} \\ R_2 - 4R_1 \\ R_3 - 5R_1 \end{matrix}$$

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & -3 & -6 & -9 \\ 0 & -5 & -10 & -15 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 4 & 5 & 6 & 7 \\ 6 & 7 & 8 & 9 \end{bmatrix} \begin{matrix} -R_1 \\ -R_2 \end{matrix}$$

1.2+3

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 3 & 3 & 3 & 3 \\ 2 & 2 & 2 & 2 \end{bmatrix}$$

$$x_1 + x_2 + x_3 = 1$$

$$x_1 + 2x_2 + 3x_3 = 9$$

$$x_2 + 2x_3 = 3; \quad x_1 = 1$$

not solvable.

$$R_2 - 4R_1 - 3 \cdot (x_2 + 2x_3) \Rightarrow 2x_2 = -2; \quad x_2 = -1$$

$$x_3 =$$

Re-calc.

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 9 & 5 & 6 & 7 \\ 6 & 2 & 8 & 9 \end{bmatrix} \begin{matrix} R_2 - 4R_1 \\ R_3 - 5R_1 \end{matrix}$$

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & -3 & -6 & -9 \\ 0 & -5 & -10 & -15 \end{bmatrix}$$

H/W 211

$$\begin{array}{ccc|c} 0 & 1 & 4 & -5 \\ 1 & 3 & 5 & -2 \\ 3 & 7 & 7 & 6 \end{array}$$

$$\begin{array}{ccc|c} 0 & -2 & -8 & 12 \\ -3 & 1 & 2 & 3 \\ 3 & 7 & 7 & 6 \\ 0 & 4 & 4 & 9 \end{array}$$

Not Solvable

Q13

$$\begin{array}{ccc|c} 1 & 0 & -3 & 0 \\ 2 & 2 & 9 & 7 \\ 0 & 1 & 5 & -2 \end{array}$$

$$R_1 + R_3 \Rightarrow \begin{array}{ccc|c} 1 & 1 & 2 & 6 \end{array}$$

$$R_2 - 2(R_1 + R_3) \Rightarrow \begin{array}{ccc|c} 0 & 0 & 5 & -5 \end{array}$$

$$x_3 = -1$$

$$x_2 = 3$$

$$x_1 = 5$$

H/W 211

$$\begin{array}{ccc|c} 0 & 1 & 4 & -5 \\ 1 & 3 & 5 & -2 \\ 3 & 7 & 7 & 6 \end{array}$$

$$\begin{array}{ccc|c} 0 & -2 & -8 & 12 \\ -3 & 1 & 2 & 3 \\ 3 & 7 & 7 & 6 \\ 0 & 4 & 4 & 9 \end{array}$$

Not Solvable

Q13

$$\begin{array}{ccc|c} 1 & 0 & -3 & 0 \\ 2 & 2 & 9 & 7 \\ 0 & 1 & 5 & -2 \end{array}$$

$$R_1 + R_3 \Rightarrow \begin{array}{ccc|c} 1 & 1 & 2 & 6 \end{array}$$

$$R_2 - 2(R_1 + R_3) \Rightarrow \begin{array}{ccc|c} 0 & 0 & 5 & -5 \end{array}$$

$$x_3 = -1$$

$$x_2 = 3$$

$$x_1 = 5$$